

Format Adherence Analysis

How closely do model responses follow the expected chain-of-thought format (paper §VI-C), and does adherence correlate with prediction correctness?

Expected Sections

- 1. Card Description
- 2. Order of Events
- 3. Synergy Analysis
- 4. Conclusion
- 5. Final Score

Adherence score: fraction of expected sections found (0–1, continuous).
Order score: are found sections in the right relative order? (0–1).
Correlation: point-biserial r between adherence and binary correctness.

gpt4o

	Value
Pairs analyzed	5,625
Accuracy	80.5% (4,529 correct / 1,096 wrong)
Mean adherence (all)	1.000 ± 0.000
Variance in adherence (all)	0.0000
Mean adherence — correct	1.000 (var=0.0000)
Mean adherence — incorrect	1.000 (var=0.0000)
Adherence delta (correct – incorrect)	+0.000

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	nan	nan	N/A
Combined score	nan	nan	N/A
Response length	-0.1786	1.543e-41	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	100.0%	100.0%	+0.0% —
Order of Events	100.0%	100.0%	+0.0% —
Synergy Analysis	100.0%	100.0%	+0.0% —
Conclusion	100.0%	100.0%	+0.0% —
Final Score	100.0%	100.0%	+0.0% —

gpt54

	Value
Pairs analyzed	5,625
Accuracy	80.1% (4,507 correct / 1,118 wrong)
Mean adherence (all)	0.313 ± 0.124
Variance in adherence (all)	0.0154
Mean adherence — correct	0.306 (var=0.0147)
Mean adherence — incorrect	0.341 (var=0.0169)
Adherence delta (correct – incorrect)	-0.035

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	-0.1134	1.437e-17	***
Combined score	-0.1211	7.678e-20	***
Response length	-0.3101	1.266e-125	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	38.1%	46.0%	-7.9% ↓
Order of Events	6.7%	17.3%	-10.5% ↓
Synergy Analysis	6.7%	5.7%	+1.0% —
Conclusion	1.3%	1.5%	-0.2% —
Final Score	100.0%	100.0%	+0.0% —

gemini10pro

	Value
Pairs analyzed	5,625
Accuracy	42.9% (2,415 correct / 3,210 wrong)
Mean adherence (all)	0.992 ± 0.047
Variance in adherence (all)	0.0022
Mean adherence — correct	0.992 (var=0.0018)
Mean adherence — incorrect	0.992 (var=0.0025)
Adherence delta (correct – incorrect)	+0.001

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	0.0091	0.4971	ns
Combined score	0.0078	0.5587	ns
Response length	-0.2236	1.079e-64	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	99.1%	99.2%	-0.1% —
Order of Events	99.1%	99.3%	-0.2% —
Synergy Analysis	98.1%	98.0%	+0.1% —
Conclusion	99.9%	99.5%	+0.4% —
Final Score	99.9%	99.7%	+0.2% —

gemini15flash

	Value
Pairs analyzed	4,810
Accuracy	75.5% (3,632 correct / 1,178 wrong)
Mean adherence (all)	1.000 ± 0.012
Variance in adherence (all)	0.0001
Mean adherence — correct	1.000 (var=0.0002)
Mean adherence — incorrect	1.000 (var=0.0000)
Adherence delta (correct – incorrect)	-0.000

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	-0.0082	0.5691	ns
Combined score	-0.0081	0.573	ns
Response length	-0.3031	9.87e-103	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	100.0%	100.0%	-0.0% —
Order of Events	100.0%	100.0%	-0.0% —
Synergy Analysis	100.0%	100.0%	-0.0% —
Conclusion	100.0%	100.0%	-0.0% —
Final Score	100.0%	100.0%	+0.0% —

gpt4omini

	Value
Pairs analyzed	5,625
Accuracy	68.1% (3,832 correct / 1,793 wrong)
Mean adherence (all)	1.000 ± 0.008
Variance in adherence (all)	0.0001
Mean adherence — correct	1.000 (var=0.0000)
Mean adherence — incorrect	1.000 (var=0.0002)
Adherence delta (correct – incorrect)	+0.000

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	0.0195	0.1438	ns
Combined score	0.0195	0.1438	ns
Response length	-0.1883	4.448e-46	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	100.0%	100.0%	+0.0% —
Order of Events	100.0%	100.0%	+0.0% —
Synergy Analysis	100.0%	99.9%	+0.1% —
Conclusion	100.0%	99.9%	+0.1% —
Final Score	100.0%	99.9%	+0.1% —

gpt4ominift

	Value
Pairs analyzed	5,625
Accuracy	79.4% (4,465 correct / 1,160 wrong)
Mean adherence (all)	1.000 ± 0.004
Variance in adherence (all)	0.0000
Mean adherence — correct	1.000 (var=0.0000)
Mean adherence — incorrect	1.000 (var=0.0000)
Adherence delta (correct – incorrect)	-0.000

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	-0.0096	0.471	ns
Combined score	-0.0105	0.4301	ns
Response length	-0.1931	2.115e-48	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	100.0%	100.0%	+0.0% —
Order of Events	100.0%	100.0%	+0.0% —
Synergy Analysis	100.0%	100.0%	-0.0% —
Conclusion	100.0%	100.0%	+0.0% —
Final Score	100.0%	100.0%	+0.0% —

gemini3flash

	Value
Pairs analyzed	5,625
Accuracy	82.3% (4,628 correct / 997 wrong)
Mean adherence (all)	0.998 ± 0.027
Variance in adherence (all)	0.0007
Mean adherence — correct	0.999 (var=0.0004)
Mean adherence — incorrect	0.994 (var=0.0022)
Adherence delta (correct – incorrect)	+0.005

Correlation with Correctness

Metric	r	p-value	Significance
Adherence score	0.0690	2.213e-07	***
Combined score	0.0800	1.902e-09	***
Response length	-0.2367	1.719e-72	***

Section Presence Rates

Section	Correct	Incorrect	Δ
Card Description	99.9%	99.1%	+0.8% —
Order of Events	99.7%	99.1%	+0.6% —
Synergy Analysis	99.9%	99.6%	+0.3% —
Conclusion	99.9%	99.1%	+0.8% —
Final Score	100.0%	100.0%	+0.0% —

Response Length Analysis

Across all models, incorrect predictions have significantly longer responses. This is a confound: models write more when pairs are harder (true synergies/anti-synergies), and harder pairs remain hard to classify regardless of response length.

Cross-Model Summary

Model	Acc	Mean (correct)	Mean (incorrect)	Diff	r	d	p (MWU)	sig
gpt4o	80.5%	1329 (σ=174)	1409 (σ=177)	-80	-0.1786	-0.458	1.15e-40	***

Model	Acc	Mean (correct)	Mean (incorrect)	Diff	r	d	p (MWU)	sig
gpt54	80.1%	804 (σ =222)	992 (σ =266)	-189	-0.3101	-0.817	1.7e-106	***
gemini10pro	42.9%	1346 (σ =293)	1486 (σ =309)	-140	-0.2236	-0.463	1.69e-72	***
gemini15flash	75.5%	1127 (σ =215)	1290 (σ =236)	-163	-0.3031	-0.739	1.41e-97	***
gpt4omini	68.1%	1375 (σ =184)	1452 (σ =189)	-76	-0.1883	-0.411	1.19e-45	***
gpt4ominift	79.4%	714 (σ =107)	768 (σ =124)	-54	-0.1931	-0.486	1.16e-42	***
gemini3flash	82.3%	1651 (σ =286)	1836 (σ =310)	-185	-0.2367	-0.638	1.71e-69	***

gpt4o — detail

Length by outcome

	Mean	Median	Std
Correct	1329	1318	174
Incorrect	1409	1397	177

- $r(\text{length, correct}) = \mathbf{-0.1786}$, $p=1.54\text{e-}41$ ***
- Cohen's d = **-0.458** (correct vs incorrect)
- $r^2 = 0.0319$ (3.19% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1412	1132	89.8%
Q2	1405	1277	82.7%
Q3	1404	1391	79.1%

Quartile	n	Mean length	Accuracy
Q4	1404	1579	70.4%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	1387	31.2%
Neutral (0)	4553	1339	83.8%
Synergy (+1)	947	1363	71.1%

gpt54 — detail

Length by outcome

	Mean	Median	Std
Correct	804	767	222
Incorrect	992	958	266

- $r(\text{length, correct}) = -0.3101$, $p=1.27\text{e-}125$ ***
- Cohen's d = **-0.817** (correct vs incorrect)
- $r^2 = 0.0961$ (9.61% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1409	573	92.8%
Q2	1409	731	86.3%
Q3	1402	886	78.4%
Q4	1405	1176	62.9%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	985	34.4%
Neutral (0)	4553	825	84.1%
Synergy (+1)	947	900	66.9%

gemini10pro — detail

Length by outcome

	Mean	Median	Std
Correct	1346	1307	293
Incorrect	1486	1444	309

- $r(\text{length, correct}) = -0.2236$, $p=1.08e-64$ ***
- Cohen's d = **-0.463** (correct vs incorrect)
- $r^2 = 0.0500$ (5.00% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1413	1080	61.1%
Q2	1403	1300	43.4%
Q3	1405	1486	38.0%
Q4	1404	1838	29.1%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	1507	27.2%
Neutral (0)	4553	1422	35.6%
Synergy (+1)	947	1432	80.1%

gemini15flash — detail

Length by outcome

	Mean	Median	Std
Correct	1127	1103	215
Incorrect	1290	1280	236

- $r(\text{length, correct}) = -0.3031$, $p=9.87\text{e-}103$ ***
- Cohen's d = **-0.739** (correct vs incorrect)
- $r^2 = 0.0918$ (9.18% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1209	905	89.7%
Q2	1197	1071	82.2%
Q3	1202	1213	76.0%
Q4	1202	1480	54.2%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	109	1284	29.4%
Neutral (0)	3892	1150	80.4%
Synergy (+1)	809	1232	58.0%

gpt4omini — detail

Length by outcome

	Mean	Median	Std
Correct	1375	1368	184
Incorrect	1452	1448	189

- $r(\text{length, correct}) = -0.1883$, $p=4.45e-46$ ***
- Cohen's d = **-0.411** (correct vs incorrect)
- $r^2 = 0.0355$ (3.55% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1412	1166	78.8%
Q2	1401	1332	71.6%
Q3	1418	1457	66.1%
Q4	1394	1646	55.9%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	1491	22.4%
Neutral (0)	4553	1401	69.1%
Synergy (+1)	947	1378	69.3%

gpt4ominift — detail

Length by outcome

	Mean	Median	Std
Correct	714	703	107
Incorrect	768	757	124

- $r(\text{length, correct}) = -0.1931$, $p=2.11e-48$ ***
- Cohen's d = **-0.486** (correct vs incorrect)
- $r^2 = 0.0373$ (3.73% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1411	593	87.2%
Q2	1409	680	83.8%
Q3	1399	750	78.0%
Q4	1406	876	68.4%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	770	16.0%
Neutral (0)	4553	720	86.8%
Synergy (+1)	947	743	52.2%

gemini3flash — detail

Length by outcome

	Mean	Median	Std
Correct	1651	1602	286
Incorrect	1836	1783	310

- $r(\text{length, correct}) = -0.2367$, $p=1.72e-72$ ***
- Cohen's d = **-0.638** (correct vs incorrect)
- $r^2 = 0.0560$ (5.60% variance explained)

Accuracy by length quartile (Q1=shortest, Q4=longest)

Quartile	n	Mean length	Accuracy
Q1	1408	1353	93.5%
Q2	1406	1553	86.3%
Q3	1405	1733	80.1%
Q4	1406	2097	69.3%

Confound check: mean length and accuracy by GT class

GT class	n	Mean length	Accuracy
Anti-syn (-1)	125	1832	48.0%
Neutral (0)	4553	1677	84.5%
Synergy (+1)	947	1698	76.0%

Length analysis generated by `length_analYsis.py`